

Software Architecture and Design Group Project

The objective of this assignment is to help students understand, apply, and evaluate software design patterns in real-world scenarios. Students are expected to design and implement a system that demonstrates proper use of multiple design patterns.

Description

Students will select **one project** from the list below and design a complete system using appropriate design patterns. The project must demonstrate the practical application of at least **three (3) design patterns**. Design patterns are not restricted to only those covered in class.

Students are required to:

- Analyze the problem
- Design the system architecture
- Implement the solution
- Justify the use of selected design patterns

Project Options (Choose ONE)

1. E-Commerce System

Design an online shopping system with product management, cart, and checkout.

2. Shape Drawing Application

Create a program that draws different shapes (Circle, Rectangle, Triangle).

3. Simple Calculator System

Build a calculator that supports multiple operations (Add, Subtract, Multiply, Divide).

4. Coffee Ordering System

Simulate a coffee shop order

5. Ride-Hailing Application

Develop a simplified ride-booking backend system.

6. Student Grading System

A system that calculates grades using different strategies.

7. User Login System

Create a login system with different authentication methods.

8. Code Refactoring Project

Refactor a poorly designed system using appropriate design patterns.

Requirements

1. Design Patterns

- Use **at least three (3)** design patterns
- Clearly explain where and why each pattern is used
- Avoid unnecessary or forced usage of patterns

2. System Design

- Provide **UML diagrams** (Class Diagram is mandatory)
- Include system architecture description

3. Implementation

- Use any programming language (Java, Python, C#, etc.)
- Code must be clean, modular, and well-documented

4. Documentation

Submit a report including:

- Problem description
- Design decisions
- Description of design patterns used
- Challenges and solutions

Marking Scheme (Total: 25 Marks)

Component	Marks
System Design (UML, Architecture)	3
Correct Use of Design Patterns	10
Code Quality and Implementation	3
Documentation and Explanation	4
Presentation / Demonstration	5

Deadline

Final Submission: After Final Exam (one day before presentation)

Presentation/Demo: One day after submission.

Submission Guidelines

Students must submit:

1. Source Code (ZIP file or repository link)
2. Documentation Report (PDF format)
3. UML Diagrams

Additional Notes

- Focus on **design quality**, not just functionality
- Clearly justify your design decisions
- Simplicity and clarity are preferred over complexity
- **if you want you can suggest your project ideas.**

Good luck, and design thoughtfully!